LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	26
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Static longitudinal stability: forces and moments Stick fixed and stick free stability, neutral point and static margin, stability curve.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Static longitudinal stability: forces and moments Stick fixed and stick free stability, neutral point and static margin, stability curve., Longitudinal dynamic stability,
5. Specific Objectives	6.1.4.1 Trainee will be able to describe aircraft stability in all the three axis.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of IB diagrams regarding CG location and Neutral Point; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Define the three axes of aircraft stability.
(b) What is the primary control surface for longitudinal stability?
(c) How does Center of Gravity (CG) position affect stability?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Forces and Moments in Longitudinal Axis • Stick Fixed vs Stick Free Stability • Neutral Point Definition and Location • Static Margin Calculation and Significance • Interpretation of Stability Curve	L & Ds	PPT, IB	15 min
(b)	• Effect of CG Shift on Pitching Moment • Relationship between Aerodynamic Center and Neutral Point • Positive vs Negative Static Margin Analysis • Stability Curve Slope and Aircraft Behavior • Comparison of Stick Fixed and Free Characteristics	L & Ds	PPT, PDF	20 min
(c)	• Dynamic Stability Implications of Static Results • Practical CG Limits for Longitudinal Stability • Summary of Stability Criteria • Link to Lateral Stability Concepts • Preparation for Next Lesson Plan	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between stick fixed and stick free stability.
(b) How is the neutral point determined relative to the CG?
(c) What does a positive static margin indicate about aircraft stability?

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of longitudinal stability discussed in this lesson apply to the lateral stability forces and moments covered in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Lateral stability: forces and Moments, effect of wing dihedral, effects of slide slip, induced roll and induced yaw. Factors effecting lateral stability. Wing sweepback effect. Dihedral effect	Refer to the master syllabus reference list	LP 27

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min